

PERSONAL SYSTEM REQUEST FLOW DIAGRAM
ASSIGNMENT 1
JASKON-Tech
ACHIVERSE PROJECT- ASAMOAH WELBECK AND
ASSOCIATES

Sarkodie Shanell

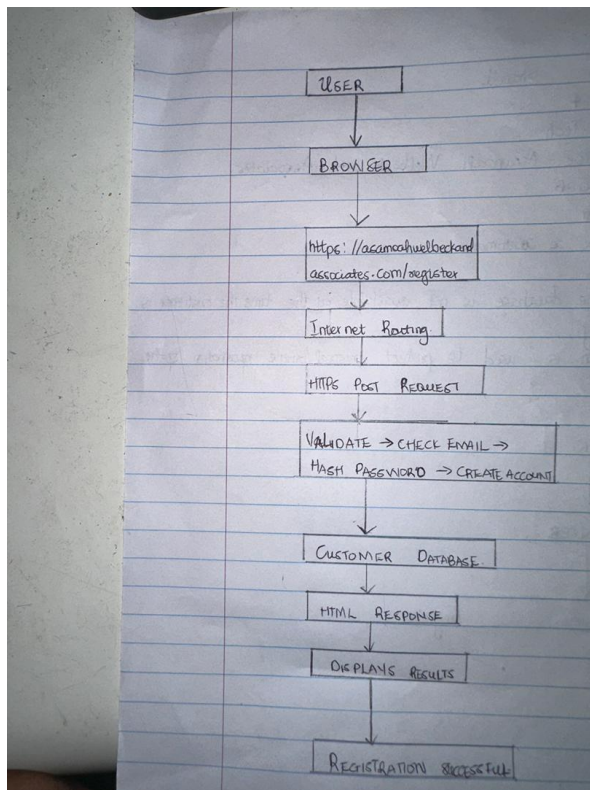
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<https://classroom.github.com/a/1-l9xPxY>

https://github.com/sarkodieshanell/personal_portfolio.git

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 Team Name: Jackson-Tech
 Project Title: ArchiVerse - Asamoah Welbeck and Associates.
 Date: 26/04/2026
 Time: 1:35pm
 Chosen Feature: Register a Customer.

Failure Point: When the database is not available at the time the customer is registering.

Security Concern: The domain is used to protect personal/private information of the user.

What device is the user on? For this request flow, the user can access the Asamoah Welbeck & associates customer registration system using either a desktop or a smartphone. In this scenario, we assume the user is using a

smartphone connected to the internet. The device allows the user to visit the registration page, enter personal information, and submit the registration form securely.

Is it a browser or mobile app? The customer registration feature is accessed through a web browser. Examples of supported browsers include Google Chrome, Microsoft Edge and Safari. The browser provides the interface where customers enter their registration details. It also sends requests to the server and displays responses returned by the application.

What URL, route, screen or endpoint is involved? The registration process begins when the user visits the Customer Registration Page using the following URL: <https://www.asamoahwelbeckandassociates.com> This page contains the registration form where customers enter information such as their full name, email address, phone number, and password. When the user clicks the **register** button, the browser sends the information to the backend API endpoint: POST <https://www.asamoahwelbeckandassociates.com/api/register>. This endpoint is responsible for receiving the registration data and creating a new customer account.

What protocol is used? The registration process uses the **HTTPS (HyperText Transfer Protocol Secure)** protocol. HTTPS encrypts all communication between the user's browser and the Asamoah Welbeck and Associates server. This prevents attackers from reading sensitive information such as passwords and personal details while the data is being transmitted across the internet. Since customer registration involves confidential information, HTTPS is essential for protecting user privacy and maintaining trust.

What data is sent? When the customer submits the registration form, the browser sends the registration information to the server. Typical data includes:

- Full Name
- Email Address
- Phone Number
- Password
- Confirm Password

This information is sent securely to the server, where it is processed before being stored in the database.

What validates the request? The request is validated by the application's business logic running on the web server. Validation ensures that all required fields have been completed correctly before a customer account is created. The system checks that the name is provided, the email address follows the correct format, the password meets the required security standards, and the password confirmation matches the original password. The application also checks whether the email address already exists in the database to prevent duplicate customer accounts.

What database table, collection or external service is touched? After successful validation, the application stores the customer's information in the Customers database table. The table contains records for each registered customer, including a unique customer ID, full name, email address, phone number, and a securely hashed password. Passwords are never stored in plain text. Instead, they are encrypted using a secure hashing algorithm before being saved. Depending on future system requirements, the application could also interact with an email service to send a registration confirmation message, but the primary storage location is the Customers table.

What response comes back? After processing the request, the server returns a response to the user's browser. If registration is successful, the response is typically returned as HTML with a success message such as **"Registration Successful."** The browser then displays a confirmation message and may automatically redirect the customer to the login page so they can sign in using their newly created account. If there is a problem with the registration, the response contains an error message explaining what needs to be corrected.

What happens if validation fails? If validation fails, the server does not create a customer account. Instead, it returns an error response to the browser describing the problem. The browser displays the error messages beside the

relevant fields, allowing the user to correct the information and submit the form again. This validation process improves data quality and prevents incorrect information from entering the database.

What happens if the database is down? If the customer database is unavailable due to maintenance or a technical failure, the server cannot save the registration information. In this situation, the application returns an error message such as **"Registration could not be completed. Please try again later."** No customer account is created until the database becomes available again. The system may also record the failure in application logs so that system administrators can investigate and restore the service as quickly as possible.

What happens if 500 users do this at once? If 500 users attempt to register simultaneously, the application should continue operating by allowing the web server to process multiple requests concurrently. The server can use request queues, load balancing, and database connection pooling to distribute the workload efficiently. If the traffic exceeds the server's capacity, some users may experience slightly slower response times. However, the application should continue accepting valid registration requests without losing customer data. Proper server scaling and performance optimization help ensure the system remains reliable during periods of high demand.

What security or privacy concern exists? The customer registration process involves sensitive personal information, including names, email addresses, phone numbers, and passwords. Therefore, strong security measures are necessary. HTTPS protects data while it is transmitted across the internet. Passwords are securely hashed before being stored in the database so they cannot be read even if the database is compromised. Input validation helps prevent malicious. Access to customer information should be restricted to authorized personnel only, and the application should comply with data protection regulations by collecting only the information necessary for registration and keeping customer data confidential.

<https://youtu.be/CvO4cwPR4XE>